CHAPTER 05

Percentages

It is one of the most important chapters which is the backbone of calculations either involved in commercial arithmetic or in real life. Personally I do maximum arithmetical calculation using percentage and others too. So in the context of calculation it is necessary to know the clear concepts of percentage which plays a very vital role in Data Interpretation besides Quantitative Aptitude section. On an average two problems i.e., nearly 4-5 % problems in QA only, are being asked in CAT every year. In other entrance/competitive exams like MAT, XAT and UPMCAT, etc there are too many questions asked from this chapter.

Chapter Checklist

- *Percentage and Its Application*
- *Percentage Conversion*
- *Percentage Change*
- *Percentage Point Change*
- *Special Cases of Percentage Change*
- *Problems Based on Population*
- ***CAT Test***

5.1 Percentage and Its Application

A fraction with denominator 100 is called a per cent. Per cent is an abbreviation for the latin word "percentum" meaning "per hundred " or "hundredhs" and is denoted by symbol %.

NOTE A fraction with denominator 10 is called as decimal.
Since per cent is a form of fraction, we can express per cent as fractions (or decimals) and *vice-versa*.

5.2 Percentage Conversion

Conversion of a Fraction into Percentage

To convert a fraction into a percentage, multiply the fraction by 100 and put "%" sign.

Exp. 1) Convert the following fractions into percentages:

(i) $\frac{1}{2}$ (ii) $\frac{3}{4}$ (iii) $\frac{4}{5}$ (iv) $\frac{7}{8}$

Solution (i) $\frac{1}{2} \to \frac{1}{2} \times 100 = 50\%$ (ii) $\frac{3}{4} \to \frac{3}{4} \times 100 = 75\%$

(iii) $\frac{4}{5} \to \frac{4}{5} \times 100 = 80\%$ (iv) $\frac{7}{8} \to \frac{7}{8} \times 100 = 87.5\%$

Conversion of a Percentage into a Fraction

To convert a percentage into a fraction, replace the % sign with $\frac{1}{100}$ and reduce the fraction to simplest form.

Exp. 2) Express the following percentage as fraction:

(i) 20% (ii) 30% (iii) 45% (iv) $5\frac{1}{8}$ % (v) 155%

Solution (i) $20\% = \frac{20}{100} = \frac{1}{5}$ (ii) $30\% = \frac{30}{100} = \frac{3}{10}$ (iii) $45\% = \frac{45}{100} = \frac{9}{20}$

(iv) $5\frac{1}{8}\% = \frac{41}{8 \times 100} = \frac{41}{800}$ (v) $155\% = \frac{155}{100} = \frac{31}{20} = 1\frac{11}{20}$

Conversion of a Percentage into a Ratio

To convert a percentage into a ratio, first convert the given percentage into a fraction in simplest form and then to a ratio.

Exp. 3) Solve the following :

(i) 38% (ii) 25% (iii) 66.66%

Solution (i) $38\% = \frac{38}{100} = \frac{19}{50} = 19:50$ (ii) $25\% = \frac{25}{100} = \frac{1}{4} = 1:4$

(iii) $66.66\% = 66\frac{2}{3}\% = \frac{200}{3 \times 100} = \frac{2}{3} = 2:3$

Conversion of a Ratio into a Percentage

To convert a ratio into a percentage, first convert the given ratio into a fraction then to a percentage.

Exp. 4) Express the following ratios as percentage :

(i) 1 : 5 (ii) 2 : 3 (iii) 4 : 9

Solution (i) $1:5 = \frac{1}{5} = \frac{1}{5} \times 100 = 20\%$ (ii) $2:3 = \frac{2}{3} = \frac{2}{3} \times 100 = 66.66\%$

(iii) $4:9 = 1/9 = 1/9 \times 100 = 44.44\%$

Conversion of a Percentage into a Decimal

To convert a percentage into a decimal remove the % sign and move the decimal point two places to the left.

Exp. 5) Convert the following percentages into decimals

(i) 36% (ii) 250% (iii) 57.5% (iv) $17\frac{1}{5}\%$ (v) 7%

Solution (i) $36\% = 0.36$ (ii) $250\% = 2.50 = 2.5$

(iii) $57.5\% = 0.575$ (iv) $17\frac{1}{5}\% = 17.2\% = 0.172$

(v) $7\% = 0.07$

Conversion of a Decimal into a Percentage

To convert a decimal into a percentage, move the decimal point two place to the right (adding zeros if necessary) and put % sign.

Exp. 6) Convert the following decimals into percentages

(i) 0.35 (ii) 8.12 (iii) 0.018

Solution (i) $0.35 = 35\%$ (ii) $8.12 = 812\%$

(iii) $0.018 = 1.8\%$

NOTE

1. Work out some more examples so that all these thing rest on your finger tips.
2. $\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \frac{4}{8} = \ldots = 50\%$ etc.

Percentage-Fraction Conversion Table

Denominators \ Numerators	1	2	3	4	5	6	7	8	9	10	11	12
1	100	200	300	400	500	600	700	800	900	1000	1100	1200
2	50	100	150	200	250	300	350	400	450	500	550	600
3	33.33	66.66	100	133.33	166.66	200	233.33	266.66	300	333.33	366.60	400
4	25	50	75	100	125	150	175	200	225	250	275	300
5	20	40	60	80	100	120	140	160	180	200	220	240
6	16.66	33.33	50	66.66	83.33	100	116.66	133.33	150	166.66	183.33	200
7	14.28	28.56	42.85	57.13	71.42	85.71	100	114.28	128.56	142.85	157.13	171.42
8	12.5	25	37.5	50	62.5	75	87.5	100	112.5	125	137.5	150
9	11.11	22.22	33.33	44.44	55.55	66.66	77.77	88.88	100	111.11	122.22	133.33
10	10	20	30	40	50	60	70	80	90	100	110	120
11	9.09	18.18	27.27	36.36	45.45	54.54	63.63	72.72	81.81	90.9	100	109.09
12	8.33	16.66	25	33.33	41.66	50	58.33	66.66	75	83.33	91.66	100
15	6.66	13.33	20	26.66	33.33	40	46.66	53.33	60	66.66	73.33	80

The Most Important Conversions

$\frac{1}{2} \to 50\%$, $\frac{1}{3} \to 33.33\%$,

$\frac{2}{3} \to 66.66\%$, $\frac{1}{4} \to 25\%$,

$\frac{3}{4} \to 75\%$, $\frac{1}{5} \to 20\%$,

$\frac{1}{6} \to 16.66\%$, $\frac{5}{6} \to 83.33\%$

The Commonly Confusing Conversions

$\frac{1}{7} = 14.28\%$ and $\frac{1}{14} = 7.14\%$

$\frac{1}{6} = 16.66\%$ and $\frac{1}{15} = 6.66\%$,

$\frac{1}{9} = 11.11\%$ and $\frac{1}{11} = 9.09\%$

$\frac{1}{15} = 6.66\%$ and $\frac{1}{16} = 6.25\%$,

$$\frac{1}{3} = 33.33\% \quad \text{and} \quad \frac{3}{10} = 30\%$$

NOTE With regard to percentage conversion, whenever you see 99.99%, you can safely assume that it is actually 100%. As, we know that $\frac{1}{9} = 11.11\%$, so we have $\frac{9}{9} = 99.99\%$ but $\frac{9}{9} = 100\%$

The Unforgettable Conversions

$$\frac{1}{13} = 7.69\%, \quad \frac{1}{17} = 5.88\%, \quad \frac{1}{19} = 5.26\%,$$
$$\frac{1}{21} = 4.76\%, \quad \frac{1}{23} = 4.35\%, \quad \frac{1}{24} = 4.166\%$$

Learnings from the Convergion Table

(i) This table is a first hand support as a percentage values of some frequently used fractions.

(ii) All the percentage values whose decimal part is 0.33, 0.66, 0.00 contain the denominator 3 in the fraction.

(iii) Similarly if there is 0.16, 0.33, 0.50, 0.66, 0.83, 0.00 it means there is a 6 in the denominator of the fraction.

(iv) Similarly if there is 0.28, 0.56, 0.85, 0.13, 0.42, 0.71 it means there is a 7 as denominator.

(v) If there is 0.11, 0.22, 0.33, 0.44 … 0.99 etc.
It means there is 9 as the denominator of the fraction.

(vi) If there is 0.09, 0.18, 0.27 … it means there is 11 as the denominator of the fraction.

(viii) If two percentage values have different decimal values (there must be different denominators) then their addition or subtraction results always in decimal *i.e.,* never as an integer.

Percentage of a Quantity

Exp. 1) Find the no. of male students (i.e., boys), if there are 42% male students in the school and the total no. of students in the school is 1000.

Solution Required number of male students

$$= 42\% \text{ of } 1000 = \frac{42}{100} \times 1000 = 420$$

Exp. 2) A students scored 85% marks in his exam. The maximum marks that any students can score in that exam are 400. How many marks did he score?

Solution Marks scored $= 85\% \text{ of } 400 = \frac{85}{100} \times 400 = 340$

Exp. 3) In an orchard $16\frac{2}{3}\%$ of the trees are mango trees.

If the total number of trees in the orchard is 360, find the number of other types of trees in the orchard.

Solution Total number of trees = 360

$$\text{Number of mango trees} = 16\frac{2}{3}\% \text{ of } 360 = \frac{50/3}{100} \times 360 = 60$$

Therefore, the number of other trees = 360 – 60 = 300

Alternatively Number of mango trees $= 16\frac{2}{3}\%$

It means no. of other types of trees

$$= \left(100 - 16\frac{2}{3}\right)\% = 83\frac{1}{3}\%$$

Thus, number of other types of trees

$$= 83\frac{1}{3}\% \text{ of } 360 = 300$$

Practice Exercise

1. Find the value of :
(i) 25% of 200 (ii) 30% of 180
(iii) 37.5% of 300 (iv) 83.33% of 480
(v) 100% of 2 quintal (vi) 165% of 330 litre
(vii) $5\frac{1}{2}\%$ of ₹ 1600 (viii) 10% of 1 hour
(ix) 66.66% of 300 (x) 20% of 1 million rupees

2. Mr. Arvind Vidyarthi spends 30% of his money on education and he has total ₹ 15000. How many rupees he spends on education?

3. Sonia purchased 80 metres of cloth, out of which 35% was used for making trousers. How much cloth was used by her for making trousers?

4. The total no. of students in a school is 1250. 40% of the students are girls. Find the number of boys.

5. William's monthly salary used to be ₹ 1140. Recently, his salary was increased by 16.66%.
(i) How much his salary (in ₹) was increased?
(ii) Find his new salary, if it was increased by 33.33%.

6. The population of Vatican city is 700. If it increases by 7.14% per annum (*i.e.,* every year). Find the population of the Vatican city after one year.

7. The usual speed of a car is 85 km/hr. If it is increased by 20%, what would be the increased speed of the car?

8. A shopkeeper announces a reduction of 8.33% on all its prices after new year. If a wrist watch was earlier for ₹ 2400. How much would it costs now?

9. 44% of the students in a class are enrolled as females and the number of students who are enrolled as males is 42. If the admitted students in that class are either males or females, find the total number of students in that class.

10. 30% of a number is 225. Find the number.

11. A horse costing ₹ 80,000 one year ago now costs 25% less. Find the changed price.

12. 1700 students took an exam 85% students passed it. Find the number of students who failed in the exam.

CAT TIPS Now I would like to suggest you that all the problems mentioned in the exercise must be done using fractions instead of percentage, in order to make the calculation simple and handy.

Answers & Solutions

1. (i) 50, (ii) 54, (iii) 112.5
(iv) 400, (v) 2 quintal (vi) 544.5
(vii) 88 (viii) 6 minutes (ix) 200
(x) 2 lakh rupees

2. $15000 \times 0.3 = 4500 \quad \left(\because 30\% = \frac{30}{100} = 0.3\right)$

Therefore, the required answer is ₹ 4500.

3. $80 \times 0.35 = 28 \quad \left(\because 35\% = \frac{30}{100} = 0.35\right)$

Therefore, the required answer is 28 meters

4. Total number of students = 1250

Total number of girls = $1250 \times 0.4 = 500$

Therefore, total number of boys in that school
$= 1250 - 500 = 750$

Therefore, the required answer is 750.

5. Since, $16.66\% = \frac{1}{6}$

So, his salary was increased by $\frac{1}{6}$ times.

So, the increase in salary $= 1140 \times \frac{1}{6} =$ ₹ 190

Similarly, $33.33\% = \frac{1}{3}$

(i) Therefore the required answer is ₹ 190.

It means his salary was increased by $\frac{1}{3}$ times.

Therefore, new salary $= 1 + \frac{1}{3} = \frac{4}{3}$ times

So, his new salary $= \frac{4}{3} \times 1140 =$ ₹ 1520.

(ii) Therefore the required answer is ₹ 1520.

6. Since $7.14\% = \frac{1}{14}$

So, the new population of the city

$$= 700\left(1 + \frac{1}{14}\right)$$

$$= 700\left(\frac{15}{14}\right) = 750$$

Therefore, the required answer is 750.

7. Since $20\% = \frac{1}{5}$

Therefore, new speed $= 1 + \frac{1}{5} = \frac{6}{5}$ times of the usual speed. That is new speed $= 85 \times \frac{6}{5} = 102$ km/hr

Therefore, the required answer is 102 km/hr

8. Since, $8.33\% = \frac{1}{12}$

So, the reduced price of the watch

$$= 2400\left(1 - \frac{1}{12}\right) = 2400\left(\frac{11}{12}\right) =$$ ₹ 2200

9. Refer the following solution

Female	Male	Total
44%	56%	100%
	42	

Since, 56% = 42, therefore, $1\% = \frac{42}{56}$

So, the $100\% = 100 \times \frac{42}{56} = 75$

Thus, the total number of students = 75

10. $30\% = 225 \Rightarrow 10\% = 75 \Rightarrow 100\% = 750$

Therefore, the required number is 750.

11. Since, $25\% = \frac{1}{4}$, therefore, $75\% = \frac{3}{4}$

So, the new cost of the horse $= \frac{3}{4}$ times of the original cost. Therefore, the new cost of the horse

$$= 80000 \times \frac{3}{4} =$$ ₹ 60000

Therefore, the required number is 60000.

12. Refer the following solution

Total	Passed	Failed
100%	85%	15%
1700		

Number of students who failed the exam
$= 15\%$ of $1700 = 0.15 \times 1700 = 255$

Therefore, the required answer is 255.

Alternatively Number of students who failed the exam = 15% of 1700

$= (10\% + 5\%) \times 1700 = (170 + 85) = 255$

Expressing One Quantity as a Percentage of Another Quantity

Exp. 1) What per cent is number 3 of number 20 ?

Solution As per cent means out of 100. Then by unitary method

$$\text{out of } 20 \to 3,\ \text{out of } 1 \to \frac{3}{20},\ \text{out of } 100 \to \frac{3}{20} \times 100 = 15\%$$

Hence to find what per cent the first number is of second number, we divide the first number by the second number and multiply the result by 100.

Exp. 2) Ravi obtained 325 marks out of a maximum of 400 marks. Find the percentage of marks obtained by him.

Solution $\text{Required percentage of marks} = \frac{325}{400} \times 100 = 81.25\%$

Exp. 3) In a factory of 150 workers, 18 were absent in a day what percentage were present?

Solution $\text{Present} = 150 - 18 = 132$

$$\text{Percentage presence} = \frac{132}{150} \times 100 = 88\%$$

Exp. 4) Kurla obtained 480 marks out of 600 and Birla obtained 560 marks out of 800.

Whose performance is better?

Solution $\%\ \text{marks of Kurla} = \frac{480}{600} \times 100 = 80\%$

$$\%\ \text{marks of Birla} = \frac{560}{800} \times 100 = 70\%$$

So, obviously Kurla's performance is better than that of Birla even though getting less absolute marks.

Introductory *Exercise 5.1*

1. What per cent is :
   (i) 30 out of 600 ?
   (ii) 25 out of 160 ?
   (iii) 75 out of 225 ?
   (iv) 36 kg of 150 kg ?
   (v) 90 cm of 4.5 metre?
   (vi) 60 litres of 40 litres?
   (vii) 800 shirts out of 1200 shirts?
   (viii) 875 m of 2 km?
2. Express:
   (i) 20 as a percentage of 500.
   (ii) 60 kg as a percentage of 80 kg .
   (iii) 350 ml as a percentage of 5.6 litre.
   (iv) ₹ 13 as a percentage of ₹ 39.
   (v) 15 seconds as a percentage of 1 hour.
   (vi) 27° as a percentage of 360°.
3. Manu scored 384 marks out of 450. What per cent marks did she get?
4. In an election, out of 60,000 eligible voters 42000 cast their vote. Calculate the percentage of eligible voters casting their votes.
5. A tin contains 24 litres of milk. Due to leakage 720 ml is lost. What per cent of milk is still present in the tin?
6. Price of an item increased from ₹ 16.50 to ₹ 41.25 Find the percentage increase in price.
7. The excise duty on a certain item has been reduced to ₹ 3480 from ₹ 5220 . Find the percentage reduction in the excise duty on that item.
8. Out of total production of 6450 tonnes of a coalmine a quantity of 645 tonnes was lost during extraction. What per cent of the total production was the net coal extracted?
9. A cricket team played 24 matches. The team won 9 matches and lost 3 matches. 12 matches ended in draw. What per cent of the total matches did the team lose?
10. In a hostel, in a particular month, out of ₹ 50,000 budget, ₹ 10,000 were allocated for the food items. Further, out of the food budget, ₹ 2000 were allocated for the fruits. What percentage of the total monthly budget of the hostel was allocated for the fruits?

5.3 Percentage Change

The application of percentage change is very diverse in nature e.g., profit and loss, simple interest and compound interest etc. All these are exactly based on the percentage increase/decrease of the original (or actual) value.

Percentage increase in a value

$$=\left(\frac{\text{increased value}-\text{original value}}{\text{original value}}\times 100\right)\%$$

Percentage decrease in a value

$$=\left(\frac{\text{original value}-\text{decreased value}}{\text{original value}}\times 100\right)\%$$

Exp. 1) Sometimes ago Abhimanyu's height was 110 cm. Now, his height is 120 cm. Find the percentage change in his height.

Solution $\frac{120-110}{110}\times 100 = 9.09\%$

Alternatively $\frac{10}{110}\to\frac{1}{11}\to 9.09\%$

Alternatively $\frac{12}{11}\to 109.09\%$, so increase $= 9.09\%$

So, the increase in height $= 109.09 - 100 = 9.09\%$

Exp. 2) The total expenses of a hostel were ₹ 8000 per month. Some students left the hostel due to which the new expenses come down by ₹ 1000. Find the percentage decrease in total expenses of the hostel.

Solution $\frac{1000}{8000}\to\frac{1}{8}\to 12.5\%$

Exp. 3) Raja's salary is ₹ 9,000 per month and Rani's salary is ₹ 10,000 per month.

(i) With respect to Raja's salary what percent is Rani's salary?

(ii) With respect to Rani's salary what percent is Raja's salary?

Solution (i) $\frac{10000}{9000}\times 100 = 111.11\%$

(ii) $\frac{9000}{10000}\times 100 = 90\%$

Alternatively (i) $\frac{10}{9}\to\frac{9}{9}+\frac{1}{9}\to 100\%+11.11\%\to 111.11\%$

(ii) $\frac{9}{10}\to 90\%$ $\left(\because \frac{1}{10}\to 10\%\right)$

5.4 Percentage Point Change

Last year Abhijeet's salary was ₹ 10,000 and Sonu's salary was ₹ 8,000. This year Abhijeet's salary is ₹ 12,000 while Sonu's salary is ₹ 10,000.

1. (i) What is the percentage increase in Abhijeet's salary?

(ii) What is the percentage increase in Sonu's salary?

(iii) Percentage increase in Sonu's salary is how much per cent greater than the percentage increase in Abhijeet's salary?

(iv) What is the percentage point change in the salary of Sonu and Abhijeet?

Solution

(i) $\frac{12{,}000-10{,}000}{10{,}000}\times 100 = 20\%$

or $\frac{2}{10}\to\frac{1}{5}\to 20\%$

(ii) $\frac{2}{8}\to\frac{1}{4}\to 25\%$

(iii) Percentage increase in Sonu's salary = 25

Percentage increase in Abhijeet's salary = 20

So the required percentage

$$=\frac{25-20}{20}\times 100 = 25\%$$

It means percentage increase in Sonu's salary is 25% greater than the percentage increase in Abhijeet's salary.

(iv) Percentage point change

= (Percentage increase in Sonu's salary

– Percentage increase in Abhijeet's salary)

= 25 – 20 = 5 percentage point

In fact, percentage point change is the difference between two percentage values.

5.5 Special Cases of Percentage Change

Getting Back the Original Value

(A) If a value p is increased by $x\%$, then we have to decrease the resultant value by $\left(\frac{x}{x+100}\times 100\right)\%$ to get back to the original value p.

$$\underset{p}{\text{Original value}} \longrightarrow \underset{\frac{p\times x}{100}}{\text{increasing value}} \longrightarrow \underset{\left(p+\frac{px}{100}\right)}{\text{Increased value}}$$

$$= p\left(\frac{100+x}{100}\right)$$

Now the percentage decrease

$$= \frac{\left[p\left(\frac{100+x}{100}\right)-p\right]}{p\left(\frac{100+x}{100}\right)}\times 100 = \left(\frac{x}{100+x}\times 100\right)\%$$

In other words (*i.e.,* in terms of fraction) if a value is increased by $\frac{n}{d}$ then to get back the same number p from the resultant value, we have to decrease the increased value by $\left(\frac{n}{d+n}\right)$.

Exp. 1) Salary of Rajeev Ratan in 2001 was \$ 100 per day and his salary in 2002 was \$ 125 per day. Again in 2003 his salary was \$ 100 per day.

(i) What is the percentage increase in his salary in 2002?

(ii) What is the percentage decrease in his salary in 2003 over 2002?

Solution (i) $\frac{125-100}{100}\times 100 = 25\%$ (ii) $\frac{125-100}{125}\times 100 = 20\%$

In fact there is same absolute change but percentage change is different due to different denominator (*i.e.,* base change). So the base change is as much important as the numerator or absolute change in quantity.

Exp. 2) Age of Ravi is 20 years. If the Shyam's age is 25% greater than that of Ravi then how much per cent Ravi's age is less than Shyam's age ?

Solution Age of Ravi Age of Shyam

$$20 \underset{(-20\%)}{\overset{(+25\%)}{\rightleftarrows}} 25$$

Solution using percentage change formula Here $x = 25$, then percentage decrease (or less) $= \frac{25}{125}\times 100 = 20\%$

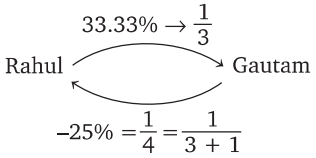
Solution using fractional change formula Since increase $= \frac{1}{4}$

$\because \quad \left(25\% = \frac{1}{4}\right)$

Therefore decrease $= \frac{1}{4+1} = \frac{1}{5} \rightarrow 20\%$

Exp. 3) Gautam has 33.33% more pencils than Rahul has. By how much per cent less pencils Rahul has than that of Gautam?

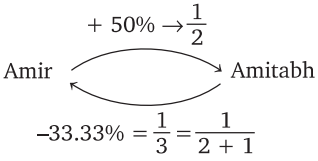
Solution



So Rahul has 25% less pencils than Gautam has.

Exp. 4) Height of Amitabh is 50% greater than the height of Amir Khan. Height of Amir Khan is how much per cent less than that of Amitabh?

Solution

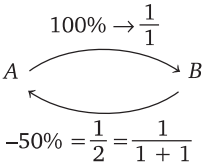


Thus the height of Amir is 33.33% less than Amitabh.

Explanation of Concept

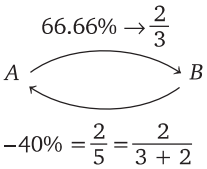
Assume A has 10 apples and B has 20 apples, then we say that B has 100% more apples than A has. Again we can say that A has 50% less apples than B has.

To calculate the percentage change :



Consider Another Example

Let us assume A has 6 chocolates and B has 10 chocolates then we can say that B has 66.66% more chocolates than A. So again we can also say that A has 40% less chocolates than B.



Remember the percentage change graphic is almost 70–80% better in giving quick and simple results. So instead of using the percentage formula, use the percentage change graphic method. For this you are just required to know the percentage fraction conversion table.

(B) If a value P is first decreased by $x\%$ then to get back the original value P, we have to increase the decreased (resultant) value by $\left(\frac{x}{100-x}\times 100\right)\%$

$$\underset{P}{\text{original value}} \xrightarrow[\frac{Px}{100}]{\text{(decreasing value)}} \underset{\left(P-\frac{Px}{100}\right)=P\left(1-\frac{x}{100}\right)}{\text{decreased value}}$$

$$\text{Now, the per cent increase} = \frac{\left[P-P\left(1-\frac{x}{100}\right)\right]}{P\left(1-\frac{x}{100}\right)}\times 100$$

$$= \frac{x}{100-x}\times 100$$

In terms of fraction, if a value P is first decreased by n/d then to get back the original number P. We have to increase the decreased (or resultant) value by $\frac{n}{d-n}$.

Exp. 5) Due to irregular working habits of Ramachandran, his salary was reduced by 20% but after some months his salary was increased to the original salary. What is the percentage increase in salary of Ramachandran?

Solution Let initial salary be ₹ 100. $100 \xrightarrow[\leftarrow]{-20\%} 80$
$\phantom{\text{Solution}}$ $\phantom{\xrightarrow[\leftarrow]{}}$ $+25\%$

Alternatively Here $x = 20$

$$\text{then \% increase} = \frac{20}{(100-20)}\times 100 = 25\%$$

Alternatively

$$\text{(Initial salary) } P \overset{-20\% \to \frac{1}{5}}{\frown} Q \text{ (Decreased salary)}$$

$$+25\% = \frac{1}{4} = \frac{1}{(5-1)} \qquad \left[\frac{n}{d-n}\right]$$

Hence his salary will be incresed by 25%.

Exp. 6) Kajol usually wears saree, which is 16.66% less than the actual length of the saree. By how much per cent the actual length of the saree is greater than the length of saree which kajol usually wears?

Solution Actual length Usual length

$$-16.66\% = \frac{1}{6}$$

$$A \quad U$$

$$+20\% = \frac{1}{5} = \left(\frac{1}{6-1}\right) \qquad \left[\frac{n}{d-n}\right]$$

So the actual length of the saree is 20% greater than the usually used saree.

NOTE

(1) If a value X is first increased by $p\%$ to Y then Y is again decreased to X by $q\%$ then p is always greater than q (for positive values).

(2) If a value X is first decreased by $p\%$ to Y and then Y is increased by $q\%$ to X, then p is always less than q.

(3) If a value A is increased by $p\%$ then again by $q\%$ once again it is increased by $r\%$, then the final value will be same as if you change the order of p, q, r *i.e.*, A can be first increased by $r\%$ then by $q\%$ and then by $p\%$ still the result will be same.

(4) The rule 3 is also applicable for the decreasing of the values. A value 'A' is first decreased by $p\%$, then by $q\%$ and then by $r\%$ and so on, the resultant value will be same as when A is first decreased by $q\%$ then by $p\%$ and then by $r\%$ etc.
Note that in case 3 and 4 we are discussing the successive increase or decrease in the value.

(5) A value 'A' is first increased by $p\%$ then by $q\%$ and then it is reduced by $r\%$ will give the same results as when A is first decreased by $r\%$, then increased by $q\%$ and then by $p\%$ etc.

Exp. 7) Initially Ms. Rakhi Sawant has ₹ 200 in her wallet then she increased it by 20%. Once again she increased her amount by 25%. The final value of money in her wallet will be how much per cent greater than the initial amount?

Solution $200 \xrightarrow{+20\%} 240 \xrightarrow{+25\%} 300$

$$\text{So the required \% increase} = \frac{300-200}{200}\times 100 = 50\%$$

Exp. 8) The age of B is 50% greater than the age of A. The age of C is 20% less than the age of B. By how much percentage the age of C is greater than the age of A?

Solution $\underset{100}{A} \xrightarrow{+50\%} \underset{150}{B} \xrightarrow{-20\%} \underset{120}{C}$

$$\text{So the required percentage change} = \frac{20}{100}\times 100 = 20\%$$

Reversing the change by same Percentage

If the value of a number is first increased by $x\%$ and then decreased by $x\%$, the net change is always a decrease (or loss) in original value. That is, % loss = $(x/10)^2\%$

If the value of a number is first decreased by $x\%$ and then the resultant value is increased by $x\%$, the net change is always a decrease (or loss) in the original value. That is % loss = $(x/10)^2$

Exp. 1) Shweta is very expert in bargaining. Once she went to a nearby shop. When Shweta asked the price of Shampoo Sachet the shopkeeper told her the price by increasing 15% of the original cost. But Shweta insisted to decrease the price by 15% so the shopkeeper sold it by decreasing the price by 15%. What is the loss or profit of shopkeeper and by how much per cent?

(a) no loss (b) profit of 1.5%
(c) loss of 2.25% (d) none of these

Solution Let the actual price be 100.

$$100 \xrightarrow{+15\%} 115 \xrightarrow{-15\%} 97.75 \quad \text{(loss of 2.25\%)}$$

Alternatively $\text{loss}\% = \left(\frac{15}{10}\right)^2 = 2.25\%$

There is always a loss.

Exp. 2) If the length and breadth of a rectangle are changed by $+20\%$ and -10%. What is the percentage change in area of rectangle?

(a) 8% (b) 10.8%
(c) 20% (d) data insufficient

Solution $l \times b = \text{area}$

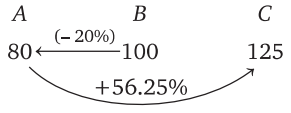
$$1 \times 1 = 1 \Rightarrow 1.2 \times 0.9 = 1.08$$

So there is 8% increase in the area of rectangle.

Exp. 3) A's monthly salary is 20% lower than B's monthly salary and C's monthly salary is 56.25% greater than A's salary. By how much percent B's salary is less than C's salary?.

(a) 15% (b) 20%
(c) 25% (d) 33.33%

Solution



(Consider the salary of $B = 100$)

The required value $= \frac{25}{125} \times 100 = 20\%$

Product Constancy

It is the same as we know the inverse proportion in the chapter of Ratio, Proportion and Variation. For example, when the rate of a pencil is ₹ 1.25 then we can purchase 16 pencils by paying ₹ 20. If the rate of a pencil is decreased by ₹ 0.25 then we can purchase 20 pencils by paying ₹ 20.

Explanation : Rate × No. of pencils = Price

$$1.25 \times 16 = 20, \quad 1.00 \times 20 = 20$$

So you can see that here the product (20) is constant in both the cases. Thus it is clear that if we reduce the price of a pencil to ₹ 0.50, then we can purchase 40 pencils in ₹ 20.

Some more examples of product constancy :

(i) speed × time = distance
(ii) rate × time = cost
(iii) efficiency × time = work
(iv) length × breadth = area
(v) average × no of elements = total value
(vi) rate × quantity = price (or expenditure)

For example The price of sugar is increased by 25% then by how much per cent should a customer reduce the consumption (*i.e.,* quantity used) of sugar so that he has not to increase his expenses on sugar.

price × quantity = expenditure

$$100 \times 100 = 10{,}000$$
$$125 \times x = 10{,}000$$
$$\Rightarrow x = \frac{10{,}000}{125} = 80. \text{ Therefore \% reduction} = 20\%$$

or
$$1 \times 1 = 1$$
$$1.25 \times k = 1 \quad \Rightarrow k = 0.8$$

Thus there will be 20% decrease in the consumption of sugar in order to maintain the same expenditure on sugar.

Product Constancy Conditions

- When one factor of a product is increased by p%, then the other factor will be decreased by $\left(\frac{p}{100+p} \times 100\right)$%.

  It means when one factor of a product is increased by $\frac{n}{d}$ then the other factor is decreased by $\frac{n}{(d+n)}$.

- When one factor of a product is decreased by p% then the other factor will be increased by $\left(\frac{p}{100-p} \times 100\right)$%.

  It means when one factor of a product is decreased by $\frac{n}{d}$ then the other factor will must be increased by $\frac{n}{(d-n)}$.

Exp. 1) If the price of a commodity be raised by 20%, then by how much per cent a house holder reduce his consumption of the same commodity so that his expenditure does not increase.

Solution Since here product (*i.e.,* expenditure) is constant

rate × consumption = expenditure

initially $\rightarrow 1 \times 1 = 1$

After change $1.2 \times x = 1$

$\Rightarrow x = 0.833$

$\therefore$ decrease in value = 16.66%

Alternatively

Increase in rate		Decrease in consumption
$+20\% = +\frac{1}{5}$	$\longrightarrow$	$-16.66\% = -\frac{1}{6}$

Exp. 2) If the price of petrol falls down by 20% by how much per cent must a person increase its consumption, so as not to decrease the expenditure on this item?

Solution Since product is constant

decrease by		increase by	
$20\% = \frac{1}{5}$	$\longrightarrow$	$\frac{1}{4} = 25\%$	$\left[\because \frac{n}{d} \rightarrow \frac{n}{(d-n)}\right]$

Exp. 3) Due to 50% increase in the price of rice . We purchased 5 kg less rice with the same amount of ₹ 60. What is the new price of rice?

(a) ₹ 4.66 (b) ₹ 5 (c) ₹ 4 (d) ₹ 6

Solution Increase in price Decrease in amount

$$50\% = \frac{1}{2} \longrightarrow \frac{1}{3} = 33.33\%$$

Since the new quantity of rice decreases by 33.33% which is equal to 5 kg it means initially there was 15 kg rice to be used.

So, the initial price = ₹ 4 $\because \left(\frac{60}{15} = 4\right)$

and final price = ₹ 6 $\because \left(\frac{60}{10} = 6\right)$

Alternatively From the options.

Let us consider choice (d).

Therefore $[6 \times 10 = 60]$ (finally)

Hence $[4 \times 15 = 60]$ (initialy)

Exp. 4) The length of a plot is decreased by 33.33% . By how much % the breadth of the plot will be increased so that the area remains constant?

Solution (Decrease) (Increase)

$$33.33 = \frac{1}{3} \longrightarrow \frac{1}{2} = 50\% \qquad \left[\frac{1}{2} = \frac{1}{3-1}\right]$$

Difference Between 'by' and 'to'

Please note that there is a clear difference between "by" and "to". For example., the income is reduced by 40% it means the new income is 60% of the original and the income is reduced to 40% means the new income is 40% of the original value. Thus "by" represents difference and "to" represents final value. For example., The income of Sarika is increased by 20% means its new income is 100 + 20 = 120 % of the original income. The income of Sarika is increased to 120% means the new income of Sarika is 120% of the original income.

Exp. 1) In an election between two candidates, the candidate who got 57% valid votes won by a majority of 420 votes. Find the total no. of valid votes .

Solution Winner Loser

$0.57x$ $0.43x$ $[100 - 57 = 43]$

$0.14x$ $[57 - 43 = 14]$

$0.14\,x = 420 \Rightarrow x = 3000$

Hence total valid votes = 3000

Exp. 2) Due to fall in manpower, the production in the factory decreases by 60%. By what per cent should the working hour be increased to restore the original production in the factory?

Solution Manpower × Working hours = Production

$$(-60\%) = \frac{3}{5} \longrightarrow \frac{3}{2} = (+150\%) \qquad \left[\frac{3}{5} \to \frac{3}{5-3} = \frac{3}{2}\right]$$

Hence by 150% working hours will be increased. It means the new working hours will be 2.5 times (not 1.5 times) of the original time.

Exp. 3) Two numbers are respectively 25% and 40% less than a third number. What per cent is the second of the first ?

Solution Consider A, B, C three numbers and assume $C = 100$ (as a base)

A B C

75 60 $\xleftarrow{(-40\%)}$ 100

$\overset{(-25\%)}{\curvearrowleft}$

Now $\frac{60}{75} \times 100 = 80\%$

Exp. 4) A person gives 10% to his wife 10% of the remaining to a hospital (as a donation) again 10% of the remaining to Prime Minister's Relief Fund. Then he has only ₹ 7290 with him. What was the initial sum of money with that person?

Solution Since he gives 10% so he is left with 90% of the original sum and since he does the same with the remaining (or left) amount. So it forms a chain.

$\therefore$ Remaining amount = $(x) \times 0.9 \times 0.9 \times 0.9 = 0.729\,x = 7290$

$\Rightarrow x = 10{,}000$, where x is supposed to be initial amount.

Exp. 5) Initially a shopkeeper had n chocolates. A customer bought 10% chocolate from n then another customer bought 20% of the remaining chocolates, after that one more customer purchased 25% of the remaining chocolates. Finally shopkeeper is left with 270 chocolates in his shop. How many chocolates were there initially in his shop?

(a) 300 (b) 450 (c) 500 (d) 600

Solution $n \times 0.9 \times 0.8 \times 0.75 = 270 \Rightarrow n = \frac{270 \times 10{,}000}{9 \times 8 \times 75} \Rightarrow n = 500$

NOTE These type of problems (see example no. 4 and 5) can be solved with a great convenience if we solve in reverse order, with the aid of given choices. Let us consider option (c)

$500 \times 0.9 = 450 \Rightarrow 450 \times 0.8 = 360 \Rightarrow 360 \times 0.75 = 270$

Hence presumed option is correct.

Difference Between Decreased value and Decreased in value

There is a huge difference between "decreased value" and "decrease in value" similarly, there is a huge between "increased value" and "increase in value".

For example Initial value = 70

Final value = 90. It means increased value = 90 but increase in value = 20 and % increase in value $= \frac{90-70}{70} \times 100 = 28.57\%$

(1) If there is an increase of x/y in any value P then the increased value will be $P\left(1 + \frac{x}{y}\right)$.

(2) If there is a decrease of x/y in any value P, then the decreased value will be $P\left(1 - \frac{x}{y}\right)$.

Exp. 1) Nishith is now 20 years old. Some years later his age will increase by 50% of himself. What will be the new age at that time?

Solution $20\left(1+\frac{1}{2}\right)=30$ years $\quad\left[\because 20+20\times\frac{1}{2}=20\left(1+\frac{1}{2}\right)\right]$

Exp. 2) The average salary of Purushottam in Infosys is 20% less than that was in Microsoft. If the salary of Purushottam in Microsoft be $ 80,000 per month then what is the salary of Purushottam in Infosys?

Solution $20\%=1/5$

So new salary $=80{,}000\left(1-\frac{1}{5}\right)=80{,}000\times\frac{4}{5}=\$\,64{,}000$

NOTE Remember this type of problems can also be solved exactly through percentages.

As $80{,}000\times0.8\ (100-20=80\%)=64{,}000$

But the technique given above is not less important. Some where it becomes very necessary to solve through fractions.

So, keep your eyes open mind focussed and use your wits to solve the problems intelligently as per the situation.

Exp. 3) Which one of the following is greatest $\frac{3}{4},\frac{7}{8},\frac{16}{19},\frac{13}{15}$?

Solution $\frac{3}{4}=75\% \quad\Rightarrow\quad \frac{7}{8}=87.5\%$

$\frac{16}{19}=84.21\% \quad\Rightarrow\quad \frac{13}{15}=86.66\%$

So, $\frac{7}{8}$ is the greatest fraction (or rational number)

5.6 Problems Based on Population

If the original population of a locality (*i.e.*, region) be P and the annual growth rate be $r\%$.

The population after n years $=P\left(1+\frac{r}{100}\right)^n$

change (or increase) in the population $=P\left[\left(1+\frac{r}{100}\right)^n-1\right]$

If there is decrease in population be $r\%$ then,

total population after n years $=P\left(1-\frac{r}{100}\right)^n$

and decrease in population $=P\left[1-\left(1-\frac{r}{100}\right)^n\right]$

Exp. 1) If the annual increase in the population be 20% and the present population be 10,000. What will be the population after 3 years hence?

(a) 16,000 (b) 17,280 (c) 14,400 (d) 1,728

Solution $10{,}000\left(1+\frac{20}{100}\right)^3=10{,}000\ \left(\frac{6}{5}\right)^3=10{,}000\times(1.2)^3=17{,}280$

Hence (b) is correct.

Exp. 2) The population of a town in the first year increases by 10% in the next year it decreases by 10%. Once again in the third year it increase by 10% and in the fourth year it decrease by 10% . If the present population be 20,000 then the population after four years will be :

(a) 16,902 (b) 19,602

(c) 20,000 (d) none of these

Solution $20{,}000\left(1+\frac{10}{100}\right)\left(1-\frac{10}{100}\right)\left(1+\frac{10}{100}\right)\left(1-\frac{10}{100}\right)$

$=20{,}000\ (1.1)\ (0.9)\ (1.1)\ (0.9)=20{,}000\times1.21\times0.81=19{,}602$

Thus option (b) is correct.

Practice Exercise

1. What per cent of $\frac{3}{7}$ is $\frac{1}{105}$?
(a) 10% (b) 2.22% (c) 45% (d) 450%

2. What per cent is 3% of 15%?
(a) 15% (b) 20% (c) 40% (d) 66.66%

3. If the cost of a calculator worth ₹ 250 is increased by ₹ 100, the rate of increase is
(a) 100% (b) 40% (c) 25% (d) 90%

4. A number increased by 37.5% gives 99, the number is
(a) 140 (b) 61.5 (c) 72 (d) 48

5. When 40% of a number is added to 42, the result is the number itself. The number is:
(a) 105 (b) 72 (c) 70 (d) 82

6. In an examination 52% of the candidates failed in Science, 42% in Mathematics and 17% in both. The no. of those who passed in both the subjects, is :
(a) 83% (b) 64% (c) 23% (d) 55.55%

7. The price of an item is increased by 20% and then decreased by 20%. The final price as compared to original price
(a) 4% more (b) 20% more (c) 20% less (d) 4% less

8. Two candidates fought an election. One got 65% of the votes and won by 600 votes. The total no. of votes polled is
(a) 12,000 (b) 10,000 (c) 2,000 (d) 8,000

9. Out of a no. of electronic items, a person purchases 60% coloured TVs 5% of these were found to be defective. The percentage of defective TVs in all is
(a) 3% (b) 6%
(c) 12% (d) Can't be determined

10. *A*'s salary is half that of *B*. If *A* got a 50% raise in his salary and *B* got a 25% raise in his salary, then the percentage increase in combined salaries of both is
(a) 30% (b) 33.33% (c) 55% (d) 28%

Answers

1. (b) **2.** (b) **3.** (b) **4.** (c) **5.** (c)
6. (c) **7.** (d) **8.** (c) **9.** (a) **10.** (b)

CAT-Test

Questions Helping you bell the CAT

LEVEL 01 > BASIC LEVEL EXERCISE

1 In our Singapore office there are 60% female employees. 50% of all the male employees are computer literate. If there are total 62% employees computer literate out of the total 1600 employees, then the no. of female employees who are computer literate, provided that our employees are either male or female.
(a) 690 (b) 672
(c) 960 (d) Can't be determined

2 The price of a car depreciates in the first year by 25% in the second year by 20% in the third year by 15% and so on. The final price of the car after 3 years, if the present cost of the car is ₹. 10,00,000 :
(a) 7,80,000 (b) 5,10,000 (c) 6,90,000 (d) 1,70,000

3 A shopkeeper charges sales tax of x% up to ₹ 2,000 and above it he charges y%.A customer pays total tax of ₹ 320, when he purchases the goods worth ₹ 6,000 and he pay's the total tax of ₹ 680 for the goods worth ₹ 12,000. The value of $(x - y)$ is :
(a) 0 (b) – 2 (c) – 4 (d) 5

4 40% of a number when added to the square of the same number, then it is increased to 4040% of itself the actual number is :
(a) 175 (b) 400 (c) 40 (d) 120

Directions (for Q. Nos. 5-7) *In my office there are 30% female employees of whom 30% earn greater than ₹ 8000 per month and 80% of male employees earn less than ₹ 8000 per month. It iw well known that my employees are either male or remale. It is well known that my employees are either male of female.*

5 What is the percentage of employees who earn more than ₹ 8000 per month?
(a) 30% (b) 23%
(c) 60% (d) Can't be determined

6 What could be the maximum percentage of employees who earn more than ₹ 8000 per month?
(a) 44% (b) 23%
(c) 60% (d) Can't be determined

7 What could be the minimum percentage of employees who earn less than ₹ 8000 per month?
(a) 9% (b) 14%
(c) 56% (d) Can't be determined

8 600 students took the test on Physics and Chemistry. 35% students failed in Physics and 45% students failed in Chemistry and 40% of those who passed in Chemistry also passed in Physics, then how many students failed in both?
(a) 162 (b) 138
(c) 60 (d) none of these

9 In an examination 70% of the candidates passed in History and 50% in Geography and 20% students failed in both the subjects. If 500 students passed in both the subjects, then how many candidates appeared for the exam ?
(a) 1,000
(b) 1,500
(c) 2,500
(d) none of the above

10 My salary is ₹ 12,345 per month. The salary of my brother is 10% greater than that of mine. The salary of my only sister is 9.09% greater than my only brother. The salary of my wife is $56\frac{12}{23}$% less than the total salary of my brother and sister together, then the salary of my wife is :
(a) greater than my sister's salary
(b) $33\frac{11}{23}$% less than my sister's salary
(c) equal to my salary
(d) $44\frac{11}{23}$% greater than my own salary

11 MDTV is a very popular TV channel. It telecasts the programmes from 8 : 00 a.m. to 12 : 00 p.m. It telecasts 60 advertisements each of 8 seconds and 16 advertisements each of 30 seconds. What is the percentage of time devoted in a day for the advertisements?
(a) 1.5% (b) 1.66%
(c) 2% (d) 2.5%

12 Lagaan is levied on the 60% of the cultivated land. The revenue department collected total ₹ 3,84,000 through the lagaan from the village of Sukhiya. Sukhiya, a very rich farmer, paid only ₹ 480 as lagaan. The percentage of total land of Sukhiya over the total taxable land of the village is :
(a) 0.15% (b) 15%
(c) 0.125% (d) none of these

13 The actual area of a rectangle is 60 cm^2, but while measuring its length a student decreases it by 20% and the breadth increases by 25%. The percentage error in area, calculated by the student is :
(a) 5% (b) 25%
(c) can't be determined (d) none of these

14 The cost of packaging of the mangoes is 40% the cost of fresh mangoes themselves. The cost of mangoes increased by 30% but the cost of packaging decreases by 50%, then the percentage change of the cost of packed mangoes, if the cost of packed mangoes is equal to the sum of the cost of fresh mangoes and cost of packaging :
(a) 14.17% (b) 7.14%
(c) 6.66% (d) none of these

15 Abhinav scores 80% in Physics and 66% in Chemistry and the maximum marks of both the papers are 100. What per cent does he score in maths which is of 200 marks, if he scores 80% marks in all the three subjects?
(a) 74% (b) 84%
(c) 87% (d) 83%

16 In the New York Stock Exchange there are 45% female employees and thus the number of male employees is exceeded by 72. Hence the total no. of employees in the New York Stock Exchange is :
(a) 540 (b) 720
(c) 7200 (d) 550

17 Three candidates *A*, *B* and *C* contested an election. Out of the total votes on a voter list 25% did not vote and 6.66% votes polled were invalid. *C* got 2450 valid votes, which were 40% more than that of *B*. If *A* got only 40% of the total votes, then who is the winner?
(a) *A* (b) B
(c) *C* (d) can't be determined

18 The cost of a car is 400% greater than the cost of a bike. If there is an increase in the cost of the car is 15% and that of bike is 20%. Then the total increase in the cost of the 5 cars and 10 bikes is :
(a) 17.5% (b) $16\frac{3}{7}\%$
(c) 18.5% (d) 18.25%

19 The square of a positive number is 2,000% greater than the number itself, then the square of that number is :
(a) 1762 (b) 1635
(c) 441 (d) 139

20 The monthly salary of Shahid and Kareena together is \$ 28,000. The salary of Shahid and Kareena is increased by 25% and 12.5% respectively then the new salary of Kareena becomes 120% of the new salary of Shahid. The new (or increased) salary of Shahid is :
(a) \$ 15,000 (b) \$ 18,000
(c) \$ 14,000 (d) \$ 16,000

21 80% of a smaller number is 4 less than 40% of a larger number. The larger number is 85 greater than the smaller one. The sum of these two numbers is :
(a) 325 (b) 425 (c) 235 (d) 500

22 220% of a number '*X*' is 44. What is 44% of '*X*' ?
(a) 88 (b) 8.8
(c) 66 (d) Data insufficient

23 The shopkeeper increased the price of a product by 25% so that customer finds it difficult to purchase the required amount. But somehow the customer managed to purchase only 70% of the required amount. What is the net difference in the expenditure on that product ?
(a) 10% more (b) 5% more
(c) 12.5% less (d) 17.5% less

24 In the previous government, party *Q* was in the opposition. Now increasing the seats by 33.33% Q is the ruling party and thus party Q enjoys twice the majority than that of party *P* in the previous government. If there were only two parties *P* and *Q* and the fix no. of seats be 500 in the parliament of Hum-Tum, then the no. of seats of the Q in the new government is :
(a) 225 (b) 200 (c) 275 (d) 300

25 In an examination a candidate got 30% marks and failed by 30 marks. If the passing marks are 60% of the total marks, then the maximum marks will be :
(a) 450 (b) 600 (c) 300 (d) 100

26 In a school there are 1800 students. Last day except 4% of the boys all the students were present in the school. Today except 5% of the girls all the students are present in the school, but in both the days no. of students present in the school, were same. The no. of girls in the school is :
(a) 1200 (b) 800 (c) 1000 (d) 600

27 In a library 60% of the books are in Hindi, 60% of the remaining are in English and rest of the books are in Urdu. If there are 3600 books in English, then the total no. of books in Urdu are :
(a) 2400 (b) 2500
(c) 3000 (d) none of these

28 In a test there are total *n* questions. Bhanu answers 20 out of 25 questions correctly in the first section. In the second section he answers 60% question correct and thus his total score is 66.66% in the test. Given that all the questions carry equal marks, without any negative marking. The total no. of question in the test is :
(a) 50 (b) 60 (c) 75 (d) 100

29 In a class of MBA students 16.66% students are from Science background and 12.5% students are from commerce background and 6.66% students from arts background and rest are from Engineering background. The minimum possible students of engineeering background are :
(a) 45 (b) 77 (c) 100 (d) 120

30 An alloy contains the copper and aluminium in the ratio of 7 : 4. While making the weapons from this alloy, 12% of the alloy got destroyed. If there is 12 kg of aluminium in the weapon, then the weight of the alloy required is :
(a) 48 kg (b) 40 kg
(c) 37.5 kg (d) 14.4 kg

31 Hariharan goes to a shop to buy an FM radio costing ₹ 1404 including sales tax at 8%. He asks the shopkeeper to reduce the price of radio so that he can save the amount equal to the sales tax. The reduction of the price of the radio is :
(a) ₹ 108 (b) ₹ 104
(c) ₹ 112.32 (d) none of these

32 The average weight of a class of students is 67.5 kg. The weight of the class teacher is 25% more than the average weight of the class. The average weight of the class is less than the class teacher by x%. The value of x is :
(a) 33.33% (b) 25%
(c) 20% (d) can't be determined

33 Last year in CAT, each section of the question paper had different weightage. The weightage of QA, DI and VA/RC sections were 8, 9 and 10 respectively. The maximum marks in all the three sections together was 810. Wrong answer did not carry negative marks as a penalty. If Padma had gotten 20% more marks in QA and 8% more marks in DI and 7.14...% more marks in VA/RC, then she must had gotten 100% marks in all the three sections. The total marks that Padma had scored :
(a) 730 (b) 700
(c) 750 (d) 775

34 A salesman gets commission on total sales at 9%. If the sale is exceeded ₹ 10,000 he gets an additional commission as bonus of 3% on the excess of sales over ₹ 10,000. If he gets total commission of ₹ 1380, then the bonus he received is :
(a) ₹ 180 (b) ₹ 120
(c) ₹ 480 (d) Data insufficient

35 In Veeru Bhai Pvt. limited company 60% of the employees are men and 48% of the employees are engineer and 66.6% of these are men. The percentage of women who are not engineers :
(a) 33.33% (b) 60%
(c) 52% (d) 46.66%

36 Initially Veer had 60% more love letters than that of Zara. In the last month the no. of love letters of Veer increased by 25% and that of Zara decreased by 25%. Again in the present month the no. of love letters of Veer decreased to 60% and that of Zara increased by 60%. Then which of the following statements is correct regarding the present no. of love letters?
(a) Veer has 40% more letters than that of Zara
(b) Zara has 20% less letters than that of Veer
(c) Veer and Zara have equal no. of letters
(d) Zara has 37.5% less letters than that of Veer.

37 The charges per hour of internet surfing is increased by 25% then find the percentage decrease in the time period of surfing of a user (a net savy) who can afford only a 10% increase in expenditure :
(a) 22% (b) 12% (c) 15% (d) 9.09%

38 The average earning of each member of the Ambani family is 20% less than the average earning of each member of the Sahara family and the total earning of Ambani's family is 20% more than the total earning of Saharas's family. The no. of family members in the Sahara is what per cent of the no. of family members of Ambani?
(a) 25% (b) 20%
(c) 66.66% (d) None of these

39 From 2000 onwards, till 2003 the price of computers increased every year by 10%. After that due to government subsidy the price of computers decreases every year by 10%. The price of a computer in 2006 will be approx. how much per cent less than the price in 2000 if the same pattern of price is continued :
(a) 2 (b) 3
(c) 4 (d) none of these

40 A book consists of 30 pages, 25 lines on each page and 35 characters on each line. If this content is written in another note book consisting of 30 lines and 28 characters per line, then the required no. of pages will how much per cent greater than the previous pages?
(a) 4.16% (b) 5%
(c) 6.66% (d) none of these

41 The rate of increase of the price of sugar is observed to be two per cent more than the inflation rate expressed in percentage. The price of sugar on January 1, 2004 is ₹ 20 per kg. The inflation rates of the years 2004 and 2005 are expected to be 8% each. The expected price of sugar on January 1, 2006 would be :
(a) ₹ 23.60 (b) ₹ 24.00 (c) ₹ 24.20 (d) ₹ 24.60

42 A club has raised 75% of the amount it needs for a new building by receiving an average donation of ₹ 600 from the people already solicited. The people already solicited represents 60% of the people the club will ask for donations. If the club is to raise exactly the amount needed for the new building, what should be the average donation from the remaining people to be solicited?
(a) 250 (b) 300 (c) 400 (d) 600

43 A number x is mistakenly divided by 10 instead of being multiplied by 10. What is the percentage error in the result?
(a) – 99% (b) + 99% (c) – 100% (d) + 100%

44 What is the percentage change in the result when we add 50 to a certain number x, instead of subtracting 50 from the same number x ?
(a) 50% (b) 100%
(c) 300% (d) can't be determined

45 In the Regional Science Centre, Lucknow the rate of ticket is increased by 50% to increase the revenue, but simultaneously 20% of the visitors decreased. What is percentage change in the revenue of Regional Science Centre. If it is known that the centre collects the revenue only from the visitors and it has no other financial supports :
(a) + 20% (b) – 25%
(c) + 30% (d) can't be determined

46 Recently when I visited a show room of shoes shopkeeper told me that he could reduce the price of Bata shoes by 49% and if I were to purchase Woodland shoes he could reduce the price to 51% of the original price. If the marked price (*i.e.,* printed price) of both the shoes was same, then which shoes was cheap to buy :
(a) Woodland (b) Bata (c) Both (d) Can't say

47 Selling price of a shirt and a coat is ₹ 4000. The cost price of a shirt is 58.33% of the cost price of a coat and so amount of profit on both the shirt and coat is same, then the price of the shirt could be :
(a) ₹ 2100 (b) ₹ 2525 (c) ₹ 2499 (d) ₹ 1120

48 On the April 1, 2005 my salary increased from ₹ 10,000 to ₹ 16,000. Simultaneously the rate of income tax decreased by 37.5% , So the amount of income tax paid by me remains constant what is the value of income tax paid by me :
(a) ₹ 3000 (b) ₹ 6000
(c) ₹ 1600 (d) Can't be determined

49 In the previous question, if the difference in the rate of income tax be 9 (in per cent) then the income tax paid by me :
(a) ₹ 2000 (b) ₹ 2400
(c) ₹ 1600 (d) none of these

50 The average of a set of whole numbers is 27.2. When the 20% of the elements (ie numbers) are eliminated from the set of numbers then the average becomes 34. The number of elements in the new set of numbers can be :
(a) 27 (b) 35 (c) 52 (d) 63

51 In a class, the no. of boys is more than the no. of girls by 12% of the total strength. The ratio of boys to girls is :
(a) 15 :11 (b) 11 : 14 (c) 14 :11 (d) 8 : 11

52 The population of a village is 5000 and it increases at the rate of 2% every year. After 2 years, the population will be :
(a) 5116 (b) 5202 (c) 5200 (d) 5204

53 A customer asks for the production of x number of goods. The company produces y number of goods daily. Out of which z% are unfit for sale. The order will be completed in :
(a) $\frac{x}{100\,y\,(1-z)}$ days (b) $\frac{100\,yz}{x}$ days
(c) $\frac{100\,x}{y\,(100-z)}$ days (d) $\frac{100}{y\,(z-1)}$ days

LEVEL 02 > HIGHER LEVEL EXERCISE

1 In the Awadh school Gomti Nagar, there are 500 students. 60% of the students are boys, 40% of whom play hockey and the girls don't play hockey, 75% of girls play badminton. There are only two games to be played. The number of students who don't play any game is :
(a) 10% (b) 36%
(c) 46% (d) can't be determined

2 A fraction in reduced form is such that when it is squared and then its numerater is increased by 25% and the denominator is reduced to 80% it results in 5/8 of the original fraction. The product of the numerator and denominator is :
(a) 6 (b) 12
(c) 10 (d) 7

3 In the Chidambaram's family the ratio of expenses to the savings is 5 : 3. But his expenses is increased by 60% and income increases by only 25% thus there is a deficit of ₹ 3500 in the savings. The increased income of Mr. Chidambaram's family is :
(a) ₹ 35,000 (b) ₹ 28,000
(c) ₹ 25,000 (d) ₹ 18,500

4 In the Presidency College two candidates contested a presidential election. 15% of the voters did not vote and 41 votes were invalid. The elected contestant got 314 votes more than the other candidate. If the elected candidate got 45% of the total eligible votes, which is equal to the no. of all the students of the college. The individual votes of each candidate are :
(a) 2250 and 1936 (b) 3568 and 3254
(c) 2442 and 2128 (d) 2457 and 2143

5 The annual earning of Mr. Sikkawala is ₹ 4 lakhs per annum for the first year of his job and his expenditure was 50%. Later on for the next 3 years his average income increases by ₹ 40,000 per annum and the saving was 40%, 30% and 20% of the income. What is the percentage of his total savings over the total expenditure if there is no any interest is applied on the savings for these four years?
(a) $49\frac{37}{87}$%
(b) $41\frac{73}{83}$%
(c) 53%
(d) none of the above

Answers

✓ *Introductory Exercise 5.1*

1. (i) 5%, (ii) 15.625, (iii) 33.33%, (iv) 24%, (v) 20%, (vi) 150%, (vii) 66.66%, (viii) 43.75%
2. (i) 4%, (ii) 75%, (iii) 6.25%, (iv) 33.33%, (v) 0.4166%, (vi) 7.5%
3. 85.33% **4.** 70% **5.** 97% **6.** 150% **7.** $33\frac{1}{3}\%$ **8.** 90% **9.** 12.5% **10.** 4%

✓ **Level 01** *Basic Level Exercise*

1 *(b)*	2 *(b)*	3 *(b)*	4 *(c)*	5 *(d)*	6 *(b)*	7 *(c)*	8 *(d)*	9 *(d)*	10 *(c)*
11 *(b)*	12 *(d)*	13 *(d)*	14 *(b)*	15 *(c)*	16 *(b)*	17 *(a)*	18 *(b)*	19 *(c)*	20 *(a)*
21 *(c)*	22 *(b)*	23 *(c)*	24 *(d)*	25 *(d)*	26 *(b)*	27 *(a)*	28 *(c)*	29 *(b)*	30 *(c)*
31 *(b)*	32 *(c)*	33 *(a)*	34 *(b)*	35 *(b)*	36 *(c)*	37 *(b)*	38 *(c)*	39 *(b)*	40 *(c)*
41 *(c)*	42 *(b)*	43 *(a)*	44 *(d)*	45 *(a)*	46 *(c)*	47 *(d)*	48 *(d)*	49 *(b)*	50 *(c)*
51 *(c)*	52 *(b)*	53 *(c)*							

✓ **Level 02** *Higher Level Exercise*

1 *(d)*	2 *(c)*	3 *(a)*	4 *(d)*	5 *(d)*	6 *(d)*	7 *(a)*	8 *(d)*	9 *(b)*	10 *(d)*
11 *(c)*	12 *(c)*	13 *(a)*	14 *(c)*	15 *(a)*	16 *(d)*	17 *(c)*	18 *(b)*	19 *(b)*	20 *(c)*
21 *(b)*	22 *(a)*	23 *(a)*	24 *(d)*	25 *(b)*	26 *(a)*	27 *(c)*	28 *(d)*	29 *(c)*	30 *(c)*
31 *(b)*	32 *(d)*	33 *(b)*	34 *(b)*	35 *(a)*	36 *(a)*	37 *(c)*	38 *(c)*	39 *(b)*	40 *(c)*
41 *(c)*	42 *(a)*	43 *(a)*	44 *(c)*	45 *(b)*	46 *(b)*	47 *(a)*	48 *(b)*	49 *(c)*	50 *(c)*

✓ **Level 03** *Final Round*

1 *(a)*	2 *(b)*	3 *(a)*	4 *(c)*	5 *(d)*	6 *(b)*	7 *(d)*	8 *(a)*	9 *(d)*	10 *(b)*
11 *(a)*	12 *(a)*	13 *(b)*	14 *(a)*	15 *(b)*	16 *(c)*	17 *(b)*	18 *(c)*	19 *(b)*	20 *(b)*
21 *(c)*	22 *(a)*	23 *(b)*	24 *(b)*	25 *(a)*	26 *(d)*	27 *(b)*	28 *(b)*	29 *(b)*	30 *(c)*
31 *(a)*	32 *(d)*	33 *(b)*	34 *(a)*	35 *(d)*	36 *(b)*	37 *(a)*	38 *(c)*	39 *(c)*	40 *(b)*
41 *(c)*	42 *(a)*	43 *(b)*	44 *(c)*	45 *(b)*	46 *(c)*	47 *(a)*	48 *(b)*	49 *(c)*	50 *(c)*

Hints & Solutions

☑ Level 01 *Basic Level Exercise*

1

F	M	
$60\,x$	$40\,x$	
$42x$	$20x$	(comp. lit.)

$\therefore$ Female comp. literate $= 1600 \times \frac{42}{100} = 672$

2 $(10,00,000) \times 0.75 \times 0.80 \times 0.85 = ₹\ 5,10,000$

3 $2000 \times \frac{x}{100} + 4000 \times \frac{y}{100} = 320$

and $2000 \times \frac{x}{100} + 10,000 \times \frac{y}{100} = 680$

$\therefore \quad x = 4 \quad \text{and} \quad y = 6$

$\therefore \quad x - y = -2$

4 Go through option $(40 \times 0.4) + (40)^2 = 1616$

$\left(100 \times \frac{1616}{40} = 4040\right)$

Alternatively $(x \times 0.4) + x^2 = \frac{x \times 4040}{100}$

$\Rightarrow \quad x^2 = 40x \quad \Rightarrow \quad x = 40$

Directions *for solutions (Q. 5-7) Let us assume that there are total number of employees at our office is 100.*

	Female	Male	Total
	30	70	100%
Salary > 8000	9	No data	No data
Salary ≤ 8000	21	No data	No data
Salary = 8000	No data	No data	No data
Salary < 8000	No data	56	No data
Salary ≥ 8000	No data	14	No data

5 Since we don't have the desired data, so we cannot determine the required value.
Hence, choice (d) is the valid one.

6 To maximize the number of employees who earn more than ₹ 8000, we can assume there is no employee who earns exactly ₹ 8000. So there would be 14 male employees and 9 female employees who earn more than ₹ 8000. That is 9 + 14 = 23

	Female	Male	Total
	30	70	100%
Salary > 8000	9	14	23
Salary ≤ 8000	21	No data	No data
Salary = 8000	No data	0	No data
Salary < 8000	No data	56	No data

Hence, choice (b) is the valid one.

7 To minimize the number of employees who earn less than ₹ 8000, we have to maximize the number of employees who earn equal to or greater than ₹ 8000. We can assume that all the 21 female employees earn exactly ₹ 8000 and so, none of them earn less than ₹ 8000. However, there are 56 male employees who definitely earn less than ₹ 8000. Therefore, minimum 56 employees must earn less than ₹ 8000.

	Female	Male	Total
	30	70	100%
Salary > 8000	9	No data	No data
Salary = 8000	21	No data	No data
Salary < 8000	0	56	56
Salary ≥ 8000	No data	14	No data

Hence, choice (c) is the valid one.

8

	Physics	Chemistry
Failed	35%	45%
Passed	65%	55%

Passed in both = 22% of total student

Percentage of students who are passed in any of the Physics or Chemistry or both = (65 + 55) – 22 = 98%

So, the percentage of students who are failed in both = 2%

Therefore total failed (in both the subject) students = 12

9

	History	Geography
Pass	$70\,x$	$50\,x$
Fail	$30\,x$	$50\,x$

both $20\,x$

$\therefore$ Total failed candidates $= (30x + 50x) - 20x = 60\,x$

Passed in both $= (100x - 60x) = 40x = 500$

Therefore total students $= 100x = 1250 \Rightarrow x = 25/2$

10 My salary = 100

Salary of my brother = 110

Salary of my sister = 120

Salary of my wife $= 230 - \left(230 \times \frac{1300}{23 \times 100}\right) = 100$

$\because \quad 56\frac{12}{23} = \frac{1300}{23}$

11 $\frac{60 \times 8 + 16 \times 30}{16 \times 60 \times 60} \times 100 = 1.66\%$

12 Total land of Sukhiya $= \frac{480\,x}{0.6} = 800\,x$

$\therefore$ Cultivated land of village $= 384000\,x$

$\therefore$ Required percentage $= \frac{800\,x}{384000} \times 100 = 0.20833\%$

13 Area $= l \times b$

$1 = 1 \times 1 \Rightarrow 1 = 0.8 \times 1.25$

So the area remained constant.

14 Cost of fresh Mangoes + Packaging cost = Total cost

$1 + 0.4 = 1.4$

$1.3 + 0.2 = 1.5$

Percentage increase in cost $= \frac{0.1}{1.4} \times 100 = 7.14\%$

15 $\frac{80}{100} + \frac{66}{100} + \frac{x}{200} = \frac{320}{400}$

$\Rightarrow \quad x = 174 \quad \Rightarrow \quad 87\%$

16

Male	Female
$55x$	$45x$

$\Rightarrow \quad 10x = 72 \Rightarrow x = 7.2$

$100x = 7.2 \times 100$

17 Let the total votes be x,

Polled votes $= 0.75x$

valid votes $= 0.70x$

$A + B + C = 0.70x$

A: $0.4x$; $B + C$: 1750; C: 2450

It means $B + C = 0.3x$ since given that $A = 0.4x$

Hence A is the winner.

18

Bike	Car
x	$5x$
$1.2x$	$5.75x$

Initially total cost $= 25x + 10x = 35x$

Changed cost $= 28.75x + 12x = 40.75x$

Percentage change $= \frac{5.75}{35} \times 100 = 16\frac{3}{7}\%$

20 Go through option

S	K
15,000	18,000
12,000	16,000

Going in the reverse direction.

Hence, the presumed option (a) is correct.

Alternatively

	Sahid	Kareena
25%	$4x$	$\frac{48x}{9}$
	$5x$	$6x$
		$+120\%$

$\Rightarrow$ Initially $\quad S : K = 4x : \frac{48x}{9} = 3 : 4$

$\Rightarrow$ Sahid's initial salary = \$12000

Sahid's changed salary = \$15000

21 Let the smaller number be x and larger number be y.

$0.8x + 4 = 0.4y$

$\Rightarrow \quad 4y - 8x = 40$

and $\quad y - x = 85 \Rightarrow x = 75$ and $y = 160$

$\therefore \quad x + y = 235$

22 $\frac{x \times 220}{100} = 44$

$\Rightarrow \quad x = 20$ So $\frac{20 \times 44}{100} = 8.8$

23 Quantity $\times$ Rate = Price

$1 \times 1 = 1$

$0.7 \times 1.25 = 0.875$

$\therefore$ Decrease in price $= \frac{0.125}{1} \times 100 = 12.5\%$

24 Go through options

	P	Q	
In the 1st government	275	225	Difference = 50
In the 2nd government	200	300	Difference = 100

Hence, the presumed option (d) is correct.

NOTE It can also be solved through the equation and variables.

25 Passing marks are $0.6x$

So $\quad 0.3x + 30 = 0.6x \quad \Rightarrow \quad x = 100$

26 Go through options

Girls	Boys
800	1000

Now, $0.95 \times 800 + 1000 = 0.96 \times 1000 + 800$

Hence, the presumed option (b) is correct .

27

H	E		U
$0.6x$	$0.24x$	$\rightarrow$	$0.16x$
	$\downarrow$		
	3600	$\longrightarrow$	2400 ($\because E : U = 24 : 16 = 3 : 2$)

28 Go through option

75 → 25, 50; 25 → 20; 50 → 30 (60%); 20, 30 → 50

$\frac{50}{75} \times 100 = 66.66\%$

Hence, the presumed option (c) is correct.

29

Science	Commerce	Arts	Engineering
$\frac{x}{6}$	$\frac{x}{8}$	$\frac{x}{15}$	K

No. of Engineering student $= 1 - \left(\frac{x}{6} + \frac{x}{8} + \frac{x}{15}\right) = k$

$= \frac{77x}{120}$

When $x = 120$ (the least possible number)

then the no of Engineering students = 77

30

Copper		Aluminium		
7	:	4		
21 kg		12 kg	→	33 kg

$$\text{Required total alloy} = \frac{33}{0.88} = 37.5\,\text{kg}$$

31 $1.08x = 1404$

$$x = 1300$$

Therefore reduction in price $= 1404 - 1300 = ₹\,104$

$$\left[\text{Since } 1300 + 1300 \times \frac{8}{100} = 1404\right]$$

So you can solve it by using options.

32 Very fundamental question.

33 $8x + 9x + 10x = 810 \Rightarrow x = 30$

Total marks in QA $\longrightarrow 240$

DI $\longrightarrow 270$

$\frac{\text{VA}}{\text{RC}} \longrightarrow 300$

Now her score in QA $\longrightarrow \frac{240}{1.2} = 200$

Her score in DI $= \frac{270}{1.08} = 250$

Her score in $\frac{\text{VA}}{\text{RC}} = \frac{300}{1.0714} = 280$

Her total score $= 200 + 250 + 280 = 730$

34 Commission up to $10000 = 10000 \times \frac{9}{100} = 900$

Again after 10000,

Commission	:	Bonus
9	:	3
$3x$	:	x

$\Rightarrow$

$\therefore$ Bonus $= (1380 - 900) \times \frac{1}{4} = ₹\,120$

35

Men	Women
$600x$	$400x$

Total Engineer $= 480\,x$

Male Engineer $= 480\,x \times 0.66 = 320\,x$

$\therefore$ Women who are Engineers $= 160\,x$

$\therefore$ Women who are not Engineers $= 400\,x - 160\,x = 240\,x$

$$\text{Required percentage} = \frac{240}{400} \times 100 = 60\%$$

36

	Veer	Zara
In the last month	160	100
After the first change	200	75
After final change	120	120

37 Time × Rate = Total charges

$1 \times 1 = 1$

$x \times 1.25 = 1.1$

$$x = \frac{1.1}{1.25} \times 100 = 88\%$$

Thus decrease in time $= 12\%$

38

	Ambani	Sahara
Average earning	$4x$	$5x$
No. of family members	k	l
Total earning	$6y$	$5y$

Average earning × Number of family members = Total earning

$$l = \frac{5y}{5x} = \frac{y}{x}$$

$$k = \frac{6y}{4x}$$

$$\text{Required percentage} = \frac{l}{k} \times 100 = \frac{\frac{y}{x}}{\frac{6y}{4x}} \times 100$$

$$= \frac{4}{6} \times 100 = 66.66\%$$

Alternatively After some steps you can use the options.

39

Year	Value
2000	100
2001	110
2002	121
2003	133.1
2004	119.79
2005	107.811
2006	97.0299

Hint $(100) \times 1.1 \times 1.1 \times 1.1 \times 0.9 \times 0.9 \times 0.9 = 97.0299$

Now, $\frac{100 - 97.0299}{100} \times 100 \approx 3\%$

40 $30 \times 25 \times 35 = x \times 30 \times 28$

$\Rightarrow$ $x = 31.25$

It means 32 pages.

So, the percentages increase in the no. of pages

$$= \frac{2}{30} \times 100 = 6.66\%$$

41 Rate of increase of the price

= (rate of inflation + 2)% = 8 + 2 = 10%

Jan 2004		Jan 2005		Jan 2006	
20	→	20 +2	→	22 +2.2	24.2

Alternatively Expected price after 2 years

$= 20 \times 1.1 \times 1.1 = 24.2$

42 Let the total number of people $= x$

then the amount donated by $0.6x$ people

$= 600 \times 0.6x = 360\,x$

Now since ₹ $360x$ is equal to 75% of the required amount. Hence we need only 25% more amount from the rest of the